STAT 410
Spring 2019
3GR, 3UG

Version A

Name ANSWERS

NetID _____

Exam 1

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Be sure to show all your work; your partial credit might depend on it.

Put your final answers at the end of your work, and mark them clearly.

If the answer is a function, specify what kind of function it is (pdf, pmf, cdf, or mgf), and its support must be included.

No credit will be given without supporting work.

The exam is closed book and closed notes.
You are allowed to use a calculator and
one 8½" x 11" sheet with notes.

Turn in all scratch paper with your exam.

Academic Integrity

The University statement on your obligation to maintain academic integrity is:

If you engage in an act of academic dishonesty, you become liable to severe disciplinary action. Such acts include cheating; falsification or invention of information or citation in an academic endeavor; helping or attempting to help others commit academic infractions; plagiarism; offering bribes, favors, or threats; academic interference; computer related infractions; and failure to comply with research regulations.

Article 1, Part 4 of the Student Code gives complete details of rules governing academic integrity for all students. You are responsible for knowing and abiding by these rules.

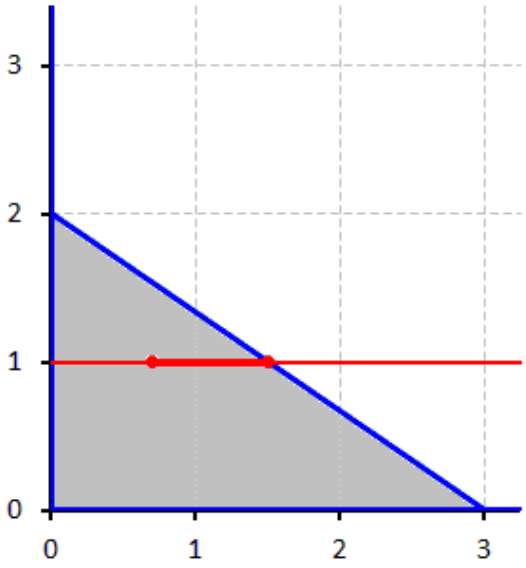
1. Every week, Alex receives 6 rubles allowance from his parents. He usually spends most of it buying candy. In Alex's favorite candy store, Skittles are sold in bulk at 2 rubles per kg, and M&M's are sold at 3 rubles per kg. Let X and Y denote the weight (in kg) of Skittles and M&M's Alex buys, respectively. Let the joint probability density function for (X, Y) be

$$f(x,y) = \frac{x+y}{5}, \qquad x>0,\ \ y>0,\ \ 2x+3y<6,$$

zero otherwise.

X – Skittles, Y – M&M's.

a) (10) Suppose we know that Alex purchased 1 kg of M&M's. What is the probability that he bought more than 0.7 kg of Skittles? That is, find $P(X>0.7 \mid Y=1)$.



$$f_Y(y) = \int_0^{3-1.5y} \frac{x+y}{5}\,dx$$

$$= \frac{x^2+2xy}{10}\Bigg|_0^{3-1.5y}$$

$$= \frac{36-12y-3y^2}{40},$$

$0<y<2.$

$$f_{X|Y}(x\mid y) = \frac{\frac{x+y}{5}}{\frac{36-12y-3y^2}{40}} = \frac{8x+8y}{36-12y-3y^2}, \qquad 0<x<3-1.5y.$$

$$f_{X|Y}(x\mid 1) = \frac{8x+8}{21}, \qquad 0<x<1.5.$$

$$P(X>0.7 \mid Y=1) = \int_{0.7}^{1.5} \frac{8x+8}{21}\,dy = \frac{4x^2+8x}{21}\Bigg|_{0.7}^{1.5} = \frac{21-7.56}{21} = \mathbf{0.64}.$$

1. (continued) $f(x,y) = \frac{x+y}{5}, \quad x>0, \ y>0, \ 2x+3y<6.$

b) (15) Find the probability distribution of the total weight of candy W = X + Y.



There are 2 cases:

$0 < w < 2, \quad 2 < w < 3.$

Technically, there are 4 cases:

$w < 0,$

$0 < w < 2, \quad 2 < w < 3,$

$w > 3,$

but $w < 0$ and $w > 3$ are boring.

$$f_W(w) = f_{X+Y}(w) = \int_{-\infty}^{\infty} f(x, w-x)\, dx. \qquad f(x, w-x) = \frac{w}{5}.$$

$x > 0$

$y > 0 \quad \Rightarrow \quad w - x > 0 \quad \Rightarrow \quad x < w$

$2x + 3y < 6 \quad \Rightarrow \quad 2x + 3w - 3x < 6 \quad \Rightarrow \quad x > 3w - 6$

$x > 0, \quad x > 3w - 6 \quad \& \quad x < w.$

Case 1: $0 < w < 2.$ $0 > 3w - 6.$

$$f_W(w) = \int_0^w \frac{w}{5}\, dx = \frac{w^2}{5}, \qquad 0 < w < 2.$$

Case 2: $2 < w < 3.$ $3w - 6 > 0.$

$$f_W(w) = \int_{3w-6}^{w} \frac{w}{5}\, dx = \frac{6w - 2w^2}{5}, \qquad 2 < w < 3.$$

OR

$$f_W(w) = f_{X+Y}(w) = \int_{-\infty}^{\infty} f(w-y,y)\,dy. \qquad f(w-y,y) = \frac{w}{5}.$$

$x > 0 \quad \Rightarrow \quad w - y > 0 \quad \Rightarrow \quad y < w$

$y > 0$

$2x + 3y < 6 \quad \Rightarrow \quad 2w - 2y + 3y < 6 \quad \Rightarrow \quad y < 6 - 2w$

$y > 0 \quad \& \quad y < w, \quad y < 6 - 2w.$

Case 1: $0 < w < 2.$ $\qquad w < 6 - 2w.$

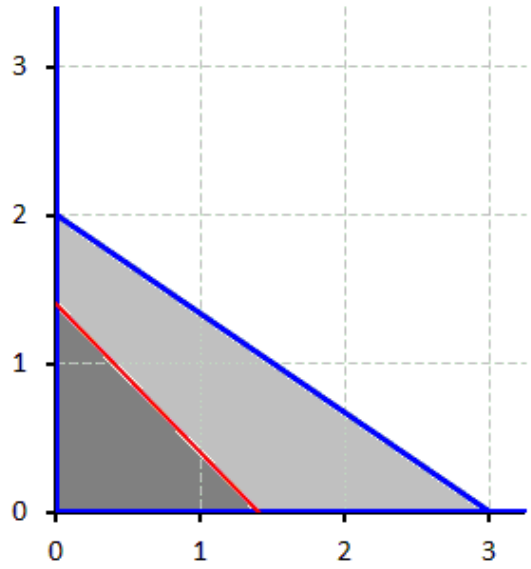
$$f_W(w) = \int_0^w \frac{w}{5}\,dy = \frac{w^2}{5}, \qquad 0 < w < 2.$$

Case 2: $2 < w < 3.$ $\qquad 6 - 2w < w.$

$$f_W(w) = \int_0^{6-2w} \frac{w}{5}\,dy = \frac{6w - 2w^2}{5}, \qquad 2 < w < 3.$$

OR

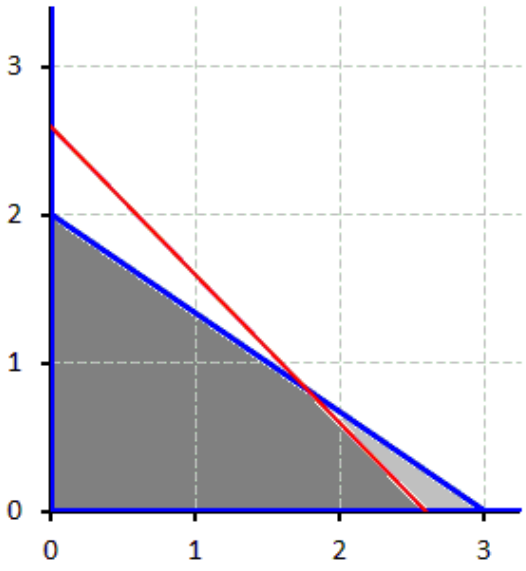
$$F_W(w) = P(X + Y \le w) = \ldots$$



Case 1: $0 < w < 2.$

$$\ldots = \int_0^w \left(\int_0^{w-x} \frac{x+y}{5}\,dy \right) dx$$

$$= \int_0^w \frac{w^2 - x^2}{10}\,dx = \frac{w^3}{15}, \qquad 0 < w < 2.$$



Case 2: $2 < w < 3.$

$2x + 3y = 6 \quad \& \quad x + y = w$

$\Rightarrow \qquad x = 3w - 6, \quad y = 6 - 2w.$

$$\ldots = 1 - \int_0^{6-2w} \left(\int_{w-y}^{\frac{6-3y}{2}} \frac{x+y}{5}\, dx \right) dy$$

$$= 1 - \int_0^{6-2w} \frac{y^2 - 12y - 4w^2 + 36}{40}\, dy$$

$$= 1 - \frac{2w^3 - 9w^2 + 27}{15}$$

$$= \frac{9w^2 - 2w^3 - 12}{15},$$

$2 < w < 3.$

OR $$\ldots = \int_0^{3w-6} \left(\int_0^{\frac{6-2x}{3}} \frac{x+y}{5}\, dy \right) dx + \int_{3w-6}^{w} \left(\int_0^{w-x} \frac{x+y}{5}\, dy \right) dx = \ldots$$

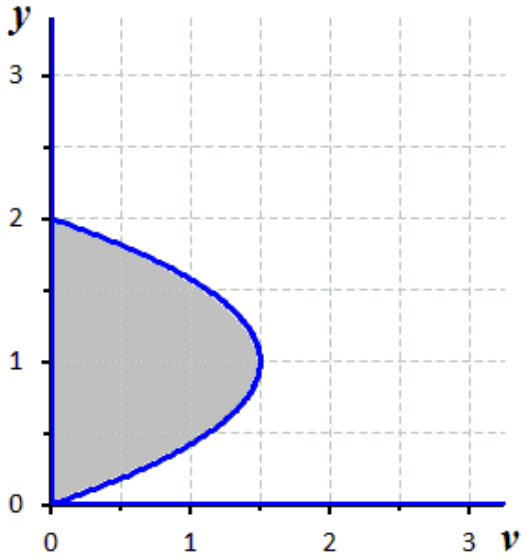
OR $$\ldots = \int_0^{6-2w} \left(\int_0^{w-y} \frac{x+y}{5}\, dx \right) dy + \int_{6-2w}^{2} \left(\int_0^{\frac{6-3y}{2}} \frac{x+y}{5}\, dx \right) dy = \ldots$$

1. (continued) $f(x,y) = \dfrac{x+y}{5}, \quad x>0,\ y>0,\ 2x+3y<6.$

c) (15) Let $V = X \cdot Y$ and $Y = Y$.

Find the joint probability density function of (V, Y), $f_{V,Y}(v,y)$.

Sketch the support of (V, Y).



$X = V/Y, \quad Y = Y.$

$y > 0$

$x > 0 \quad \Rightarrow \quad \dfrac{v}{y} > 0 \quad \Rightarrow \quad v > 0$

$2x + 3y < 6 \quad \Rightarrow \quad 2\dfrac{v}{y} + 3y < 6 \quad \Rightarrow \quad 2v < 6y - 3y^2 = 3y(2-y)$

$\Rightarrow \quad v < 1.5y(2-y)$

$$J = \begin{vmatrix} \dfrac{1}{y} & -\dfrac{v}{y^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{y}.$$

$$f_{V,Y}(v,y) = f_{X,Y}\left(\frac{v}{y}, y\right) \times |J| = \frac{\frac{v}{y}+y}{5} \times \frac{1}{y} = \frac{v+y^2}{5y^2} = \frac{v}{5y^2} + \frac{1}{5},$$

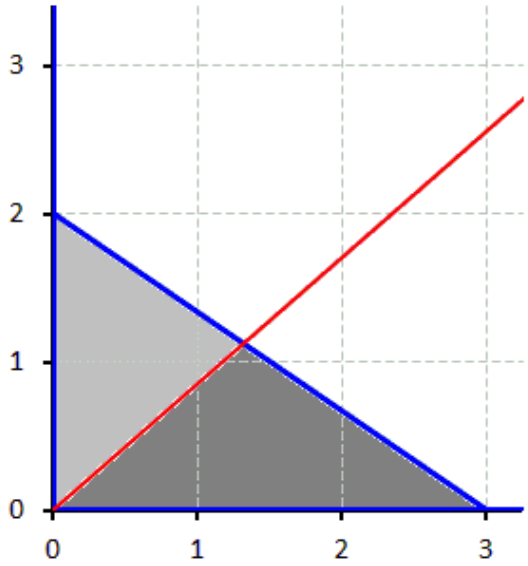
$$0 < y < 2, \quad 0 < v < 1.5y(2-y).$$

1. (continued) $f(x,y) = \dfrac{x+y}{5}, \quad x>0,\ y>0,\ 2x+3y<6.$

d) (8) Let $U = \dfrac{Y}{X} = \dfrac{\text{M\&M's}}{\text{Skittles}}$.

Set up the integral(s) for the c.d.f. of U, $F_U(u)$ OR for the p.d.f. of U, $f_U(u)$.

You do NOT have to evaluate the integral(s).



$$F_U(u) = P\left(\frac{Y}{X} \le u\right) = P(Y \le uX) = \ldots$$

$$2x+3y=6 \ \ \& \ \ y=ux$$

$$\Rightarrow \quad 2x+3ux=6$$

$$\Rightarrow \quad x=\frac{6}{2+3u}, \quad y=\frac{6u}{2+3u}.$$

$$\ldots = \int_0^{\frac{6u}{2+3u}} \left(\int_{\frac{y}{u}}^{\frac{6-3y}{2}} \frac{x+y}{5}\,dx \right) dy$$

OR $$\ldots = \int_0^{\frac{6}{2+3u}} \left(\int_0^{ux} \frac{x+y}{5}\,dy \right) dx + \int_{\frac{6}{2+3u}}^{3} \left(\int_0^{\frac{6-2x}{3}} \frac{x+y}{5}\,dy \right) dx$$

OR $$\ldots = 1 - \int_0^{\frac{6}{2+3u}} \left(\int_{ux}^{\frac{6-2x}{3}} \frac{x+y}{5}\,dy \right) dx$$

OR

Let $\mathrm{X} = \mathrm{X}, \quad \mathrm{U} = \dfrac{\mathrm{Y}}{\mathrm{X}}.$ Then $\mathrm{X} = \mathrm{X}, \quad \mathrm{Y} = \mathrm{U}\,\mathrm{X}.$

$$\mathrm{J} = \begin{vmatrix} 1 & 0 \\ u & x \end{vmatrix} = x.$$

Then $f_{\mathrm{X,U}}(x,u) = f_{\mathrm{X,Y}}(x,u\,x)\,|x| = \dfrac{x+u\,x}{5}\cdot x = \dfrac{1+u}{5}\cdot x^2.$

$x > 0,$

$y > 0 \quad \Rightarrow \quad u\,x > 0 \quad \Rightarrow \quad u > 0,$

$2x + 3y < 6 \quad \Rightarrow \quad 2x + 3\,u\,x < 6 \quad \Rightarrow \quad x < \dfrac{6}{2+3\,u}.$

$$f_{\mathrm{U}}(u) = \int_{-\infty}^{\infty} f_{\mathrm{X,U}}(x,u)\,dx = \int_{0}^{\frac{6}{2+3\,u}} \frac{1+u}{5}\cdot x^2\,dx = \frac{1+u}{15}\cdot\left(\frac{6}{2+3\,u}\right)^3, \quad u > 0.$$

OR

Let $\mathrm{Y} = \mathrm{Y}, \quad \mathrm{U} = \dfrac{\mathrm{Y}}{\mathrm{X}}.$ Then $\mathrm{X} = \dfrac{\mathrm{Y}}{\mathrm{U}}, \quad \mathrm{Y} = \mathrm{Y}.$

$$\mathrm{J} = \begin{vmatrix} -\dfrac{y}{u^2} & \dfrac{1}{u} \\ 0 & 1 \end{vmatrix} = -\frac{y}{u^2}.$$

Then $f_{\mathrm{Y,U}}(y,u) = f_{\mathrm{X,Y}}\left(\dfrac{y}{u}, y\right)\left|-\dfrac{y}{u^2}\right| = \dfrac{\frac{y}{u}+y}{5}\cdot\dfrac{y}{u^2} = \dfrac{1+u}{5\,u^3}\cdot y^2.$

$y > 0,$

$x > 0 \quad \Rightarrow \quad \dfrac{y}{u} > 0 \quad \Rightarrow \quad u > 0,$

$2x + 3y < 6 \quad \Rightarrow \quad 2\,\dfrac{y}{u} + 3y < 6 \quad \Rightarrow \quad y < \dfrac{6\,u}{2+3\,u}.$

$$f_{\mathrm{U}}(u) = \int_{-\infty}^{\infty} f_{\mathrm{Y,U}}(y,u)\,dy = \int_{0}^{\frac{6\,u}{2+3\,u}} \frac{1+u}{5\,u^3}\cdot y^2\,dy = \frac{1+u}{15\,u^3}\cdot\left(\frac{6\,u}{2+3\,u}\right)^3, \quad u > 0.$$

2. Let X_1, X_2, X_3, X_4 be a random sample (i.i.d.) of size $n = 4$ from a continuous probability distribution with the probability density function

$$f_X(x) = \frac{x}{30}, \qquad 2 < x < 8, \qquad \text{zero otherwise.}$$

a) (7) Find $P(\min X_i > 4)$.

$$F_X(x) = P(X \le x) = \int_2^x \frac{u}{30}\, du = \frac{u^2}{60}\Big|_2^x = \frac{x^2 - 4}{60}, \qquad 2 < x < 8.$$

$$P(\min X_i > 4) = [P(X > 4)]^4 = [1 - F_X(4)]^4$$

$$= \left[1 - \frac{4^2 - 4}{60}\right]^4 = 0.80^4 = \mathbf{0.4096}.$$

b) (7) Find $P(\max X_i > 7)$.

$$P(\max X_i > 7) = 1 - P(\max X_i \le 7) = 1 - [P(X \le 7)]^4$$

$$= 1 - [F_X(7)]^4 = 1 - \left[\frac{7^2 - 4}{60}\right]^4 = 1 - [0.75]^4 \approx \mathbf{0.6836}.$$

3. Consider a mixed random variable X with

the p.m.f. of the discrete portion of the probability distribution

$$p(2) = c, \qquad p(8) = 0.10, \qquad \text{zero otherwise,}$$

and the p.d.f. of the continuous portion of the probability distribution

$$f(x) = \frac{x}{40}, \qquad 2 < x < 8, \qquad \text{zero otherwise.}$$

a) (6) Find the value of c that would make this a valid probability distribution.

$$1 = \sum_{\text{all } x} p(x) + \int_{-\infty}^{\infty} f(x)\,dx$$

$$= [\,c + 0.10\,] + \int_{2}^{8} \frac{x}{40}\,dx = c + 0.10 + \frac{x^2}{80}\Big|_{2}^{8} = c + 0.10 + 0.75.$$

$$\Rightarrow \qquad c = \mathbf{0.15}.$$

b) (7) Find $\mathrm{E}(\mathrm{X})$.

$$\mathrm{E}(\mathrm{X}) = \sum_{\text{all } x} x \cdot p(x) + \int_{-\infty}^{\infty} x \cdot f(x)\,dx$$

$$= 2 \cdot 0.15 + 8 \cdot 0.10 + \int_{2}^{8} x \cdot \frac{x}{40}\,dx = 0.30 + 0.80 + \frac{x^3}{120}\Big|_{2}^{8}$$

$$= 1.1 + 4.2 = \mathbf{5.3}.$$